

Dated:

Assignment #01

Q# 1(a)

$$\text{Physical} = \text{Segment} \times 10h + \text{Offset}$$

$$\begin{aligned} * 4A5C : 0007 &\rightarrow 4A5C0h + 0007h = 4A5C7h \\ * 4A00 : 05C7 &\rightarrow 4A000h + 05C7h = 4A5C7h \end{aligned}$$

(b)

Logical address is the two 16-bit values the assembler works with. Physical address is the actual 20-bit value on the ~~box~~ bus. They are combined during execution by BIU which shifts segment left 4 bits and adds offset.

Q # 2

- 1) AX $\rightarrow$ B
- 2) BX $\rightarrow$ ~~B~~E
- 3) CX $\rightarrow$ H
- 4) DX $\rightarrow$ F
- 5) CS $\rightarrow$ C
- 6) DS $\rightarrow$ G
- 7) SS $\rightarrow$ A
- 8) ES $\rightarrow$ D

Dated:

Q#3

a) $003A34F00\text{ h}$

b) $004FA300\text{ h}$

c) byte order is used to determine which is ~~used~~ most/least significant.

Q#4

$$\begin{array}{r} 05F0 \\ + 0A20 \\ \hline 1010 \end{array}$$

$$\begin{array}{r} 0002 \\ - 0005 \\ \hline FFFD \end{array}$$

$$\begin{array}{r} 1010 \\ + 1 \\ \hline 1011 \end{array}$$

$$\begin{array}{r} FFFD \\ - 1 \\ \hline FFFC \end{array}$$

$$AX = 05F0\text{ h}$$

$$AX = 1010\text{ h}$$

$$BX = 0002\text{ h}$$

$$BX = FFFD\text{ h}$$

$$AX = 1011\text{ h}$$

$$BX = FFFC\text{ h}$$

$$CF = - \quad SF = -$$

$$CF = 0 \quad SF = 0$$

$$CF = - \quad SF = -$$

$$CF = 1 \quad SF = 1$$

$$CF = 0 \quad SF = 0$$

$$CF = 1 \quad SF = 1$$

Dated:

Q # 5 (a)

MOV AX, BookPrice

~~MOV~~ ADD AX, BookPrice

ADD AX, BookPrice

ADD AX, BookPrice

ADD AX, BookPrice

(b)

SUB AX, Discount

(c)

650 = 0282 h

5000 h

5001 h

82 h
02 h

(d)

MOV AX, CashPaid

SUB AX, FinalBill

* AX = 350

(e)

with QTY = 9, every change to QTY means rewriting the instruction count. A MUL instruction is much better.

Dated:

Q #6

* SCORES DB 6 DUP(0)

* MAX VAL DW 9999

* NAME DB "COMSATS", 0

* BUFFER DD 4 DUP(?)

Q #7

#	TYPE	Translation
1)	directive	no
2)	dix	no
3)	dix	no
4)	dix	no
5)	dix	no
6)	instruction	no
7)	inst	yes
8)	inst	yes
9)	inst	yes
10)	inst	yes
11)	dix	no
12)	dix	no

Dated:

Q# 8

- I) $1\text{MB}/64\text{KB} = 16$ chips
 $\log_2(16) = 4$ address lines
- II) * device drivers/bootloaders needing exact timing and hardware control
* performance critical inner loops that need precise control over registers
- III) A program runs unmodified across different hardware/platforms. Assembly is non-portable because its instructions map directly to one CPU's family's registers.
- IV) Each abstraction layer acts as a VM for the layer above it, hiding the details of the layer below.
- V) A linker combines object modules and libraries into one executable and resolves references. Even a single file program needs it to link in runtime libraries.
- VI) Real mode: 20 bit segmented addressing, no protection
Protected mode: 32 bit addressing, virtual memory
- VII) If an identifier could start with a digit, the assembler could not tell it apart from a numeric literal.